

Hemchand Lingamaneni

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SUMMARY

Software Engineer with experience in building secure, scalable applications across full-stack, cloud, and AI/ML domains. Skilled in designing microservices architectures, implementing role-based access control, and developing real-time messaging systems using WebSockets and Kafka. Proficient in deploying containerized applications on AWS, Azure, and GCP with Docker and Kubernetes, and automating workflows with Airflow and Terraform. Strong background in database optimization (PostgreSQL, MySQL, MongoDB) and data-driven features, including RAG pipelines with Hugging Face embeddings and LLaMA-3 for personalized recommendations. Adept at improving system reliability, scalability, and user engagement in Agile environments. Collaborates effectively with cross-functional teams to deliver scalable, maintainable code in fast-paced environments.

SKILLS

Programming Languages: Java, Python, C++, JavaScript, TypeScript, GO, SQL

Frameworks: Spring, React.js, Node.js, Angular, Next.js

Web Technologies: HTML, CSS, Bootstrap, jQuery, Ajax, JSON, XML

Databases: MySQL, SQL Server, PostgreSQL, Oracle, MongoDB, Firebase

Cloud & DevOps Tools: AWS, Azure, GCP, Docker, Kubernetes, Terraform

Big Data & ML: Kafka, Spark, Airflow, TensorFlow, PyTorch, Scikit-learn, XGBoost, Hugging Face, Llama-3, LangChain

Version Control/Other Tools: Git, GitHub, Maven, Gradle, Jira, Jenkins, Junit, JMeter

EXPERIENCE

Software Engineer | Lam Research, USA

May 2025 – Current

- Delivered new frontend features in React.js and TypeScript with Tailwind CSS, enabling engineers to track wafer lots, chamber status, and process alarms in real time.
- Built and deployed microservices with Node.js (TypeScript) on Kubernetes, integrating with equipment telemetry APIs to support tool health monitoring.
- Implemented secure access controls with OAuth2 and SSO, ensuring role-based access for fab engineers and technicians.
- Contributed to streaming pipeline using Apache Kafka, normalizing equipment event data for dashboards and analytics, reduced latency of live tool updates by 15%.

Software Engineer | Frugal Innovation Hub, USA

Aug 2023 – April 2024

- Designed a scalable access control system using Node.js and JWT, enforcing fine-grained authorization policies and ensuring secure, role-specific user experiences for parents and tutors.
- Implemented a responsive React.js interface with component-level state management and optimized rendering, leading to smoother navigation and longer session durations.
- Built a real-time messaging system using WebSockets and Kafka, ensuring instant communication with 99.9% message delivery reliability.
- Engineered a RAG recommendation system with MongoDB Atlas Vector Search and Hugging Face embeddings to analyze student assignments and chats, identifying weak concepts and surfacing targeted feedback.
- Integrated the LLaMA-3 Instruct model with the RAG workflow to generate personalized study plans on tutor and parent dashboards, improving student performance.
- Developed a centralized Admin application using Firebase to manage users, tutors, parents, and students, reducing administrative overhead through real-time synchronization and streamlined workflows.
- Deployed containerized services on AWS EC2 with Docker and S3, adding load balancing and CloudWatch monitoring to ensure 99.9% uptime and scalable performance.

Full Stack Developer | Amaravathi Software Innovations, India

May 2020 - Aug 2022

- Built a scalable e-commerce platform by integrating a Spring Boot backend with a React.js frontend, optimizing rendering and API calls to reduce page response times by 40%.
- Designed RESTful APIs for catalog, authentication, cart, and orders, integrating Razorpay for secure payments and third-party logistics for end-to-end order fulfillment.
- Optimized MySQL queries through indexing and schema restructuring, significantly reducing product search and analytics latency and improving backend responsiveness.
- Migrated a monolithic system to Dockerized microservices on AWS EKS, leveraging Kubernetes Deployments, Services, ALB Ingress, and HPA to reduce infrastructure costs by 20% and enhance scalability and fault tolerance.
- Developed Python-based ETL pipelines extracting data from MySQL into AWS S3, and automated workflows with Apache Airflow to deliver sales, inventory, and user dashboards, cutting manual prep by 60%.

EDUCATION

Master's in Computer Science and Engineering | Santa Clara University, Santa Clara, CA

Jun 2024

Coursework: Design and Analysis of Algorithms, Databases, Machine Learning, Artificial Intelligence, Networking

Bachelor's in Computer Science and Engineering | PES University, Bangalore, India

May 2022

Coursework: Web Development, Big Data, Distributed Systems, Cloud Computing, Operating Systems, Computer Networks